



# Keshav Memorial Engineering College

A unit of Keshav Memorial Technical Educational Society (KMTES)

(Approved by AICTE, New Delhi & Affiliated to Osmania University, Hyderabad)

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CIE-I

B.E (CSE) I Sem.

Set 2

## Sub: Programming for Problem Solving

Time: 12.00 PM-1.00 PM - AN

Date: 23-06-2026

Max. Marks: 20

Note:

1. First question is compulsory which consists of 6 short answers each carrying 1 mark.
2. Answer any two from the remaining three questions each carrying 7 marks.
3. Answer to each question must be written in one place only.

| Q. No. | Questions                                                           | Marks | CO  | BT  |
|--------|---------------------------------------------------------------------|-------|-----|-----|
| 1.     | (a) Describe stack and queue.                                       | (1)   | [1] | [2] |
|        | (b) Does C support polymorphism?                                    | (1)   | [3] | [4] |
|        | (c) Define abstraction.                                             | (1)   | [1] | [1] |
|        | (d) Write a short note on data encapsulation.                       | (1)   | [4] | [1] |
|        | (e) What are the uses of recursion?                                 | (1)   | [3] | [3] |
|        | (f) Define object oriented programming.                             | (1)   | [1] | [1] |
| 2.     | (a) Explain the concept of inheritance and its real life examples.  | (4)   | [1] | [2] |
|        | (b) What is a class is a class? How is it different from an object? | (3)   | [1] | [1] |
| 3.     | (a) How does virtual memory work in the operating system?           | (3)   | [1] | [1] |
|        | (b) Purpose of garbage collection in Java.                          | (4)   | [1] | [1] |
| 4.     | (a) Advantages and disadvantages of cloud computing.                | (5)   | [1] | [6] |
|        | (b) Can you explain time complexity using big O notation?           | (2)   | [1] | [2] |